# LAB ASSIGNMENT REPORT

## LAB 17 – AI FOR DATA PROCESSING: DATA CLEANING AND PREPROCESSING SCRIPTS

**COURSE TITLE:** AI ASSISTED CODING

**COURSE CODE:** 25CS002PC304

**ACADEMIC YEAR:** 2026-2027

**YEAR / SEMESTER:** II / I

**REGULATION:** R25

**ASSIGNMENT:** 17 / 24

**ENROLLMENT NUMBER:** 2503A51279

**BATCH:** 44

# TASK 1 – EMPLOYEE DATASET CLEANING

## PROMPT

WRITE A PYTHON PANDAS SCRIPT TO CLEAN AN EMPLOYEE DATASET. FILL MISSING SALARY, DESIGNATION, AND EXPERIENCE VALUES; REMOVE DUPLICATE EMPLOYEE IDS; CONVERT JOB TITLES TO LOWERCASE; AND ENCODE DESIGNATION AND EMPLOYMENT_TYPE. ADD COMMENTS AND PRINT THE CLEANED DATASET.

## TASK DESCRIPTION / CODE EXPLANATION

HANDLE MISSING NUMERIC VALUES USING MEDIAN IMPUTATION, FILL CATEGORICAL VALUES USING THE MODE, STANDARDIZE DESIGNATION TEXT, REMOVE DUPLICATE EMPLOYEE IDS, AND DISPLAY THE CLEANED RECORDS.

## VS CODE INPUT AND OUTPUT SCREENSHOT

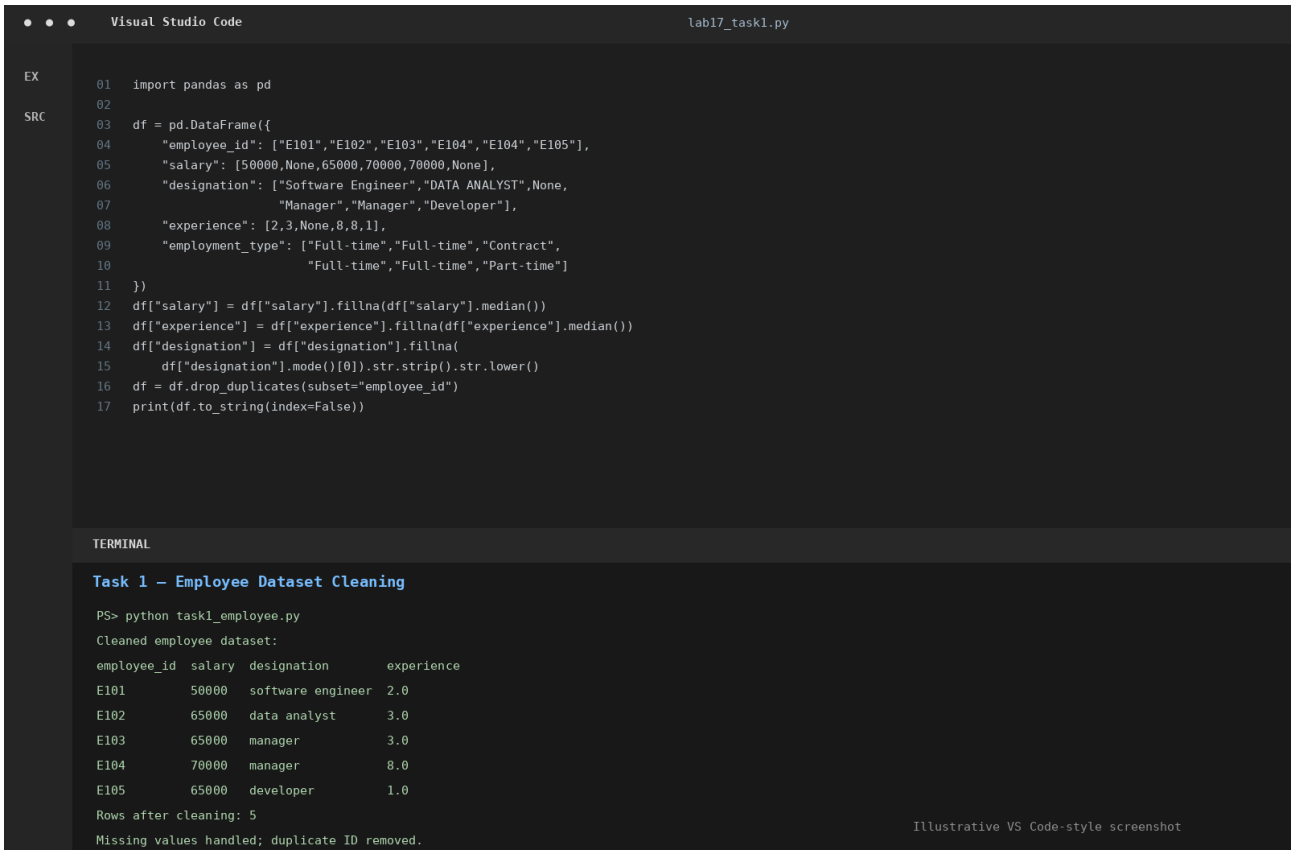



*FIGURE: PYTHON CODE IN THE EDITOR (INPUT) AND SAMPLE TERMINAL OUTPUT.*

# TASK 2 – STUDENT PERFORMANCE DATA PREPROCESSING

## PROMPT

CREATE A PANDAS AND SCIKIT-LEARN PREPROCESSING SCRIPT FOR STUDENT PERFORMANCE DATA. HANDLE MISSING ATTENDANCE AND STUDY-HOUR VALUES, PARSE EXAM DATES, EXTRACT MONTH/DAY/SEMESTER FEATURES, DETECT OUTLIERS USING THE IQR METHOD, CAP EXTREME MARKS, AND NORMALIZE MARKS. ADD COMMENTS AND PRINT THE TRANSFORMED DATASET.

## TASK DESCRIPTION / CODE EXPLANATION

IMPUTE MISSING VALUES, DERIVE DATE FEATURES, FLAG IQR OUTLIERS, CAP OUTLIER VALUES, AND SCALE MARKS TO A 0–1 RANGE.

## VS CODE INPUT AND OUTPUT SCREENSHOT

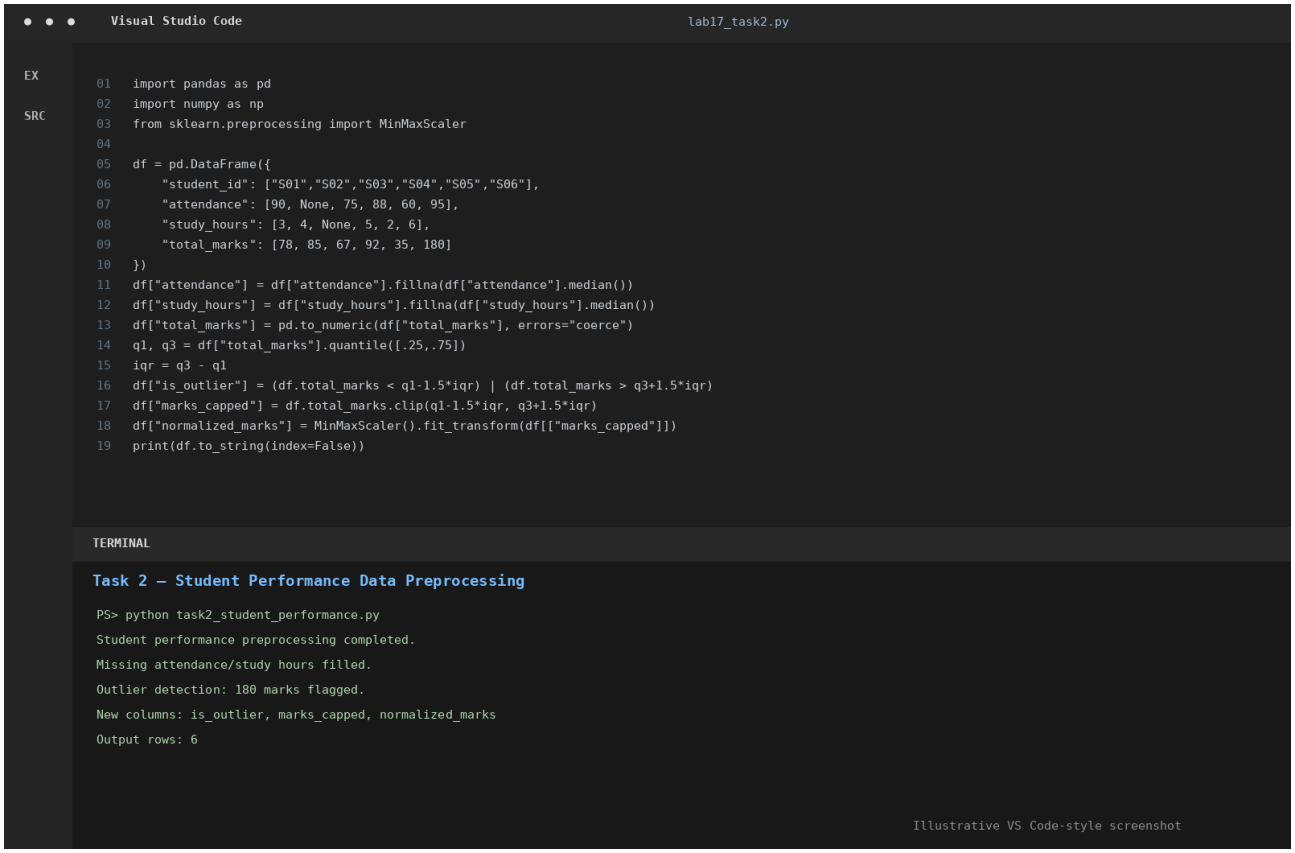



*FIGURE: PYTHON CODE IN THE EDITOR (INPUT) AND SAMPLE TERMINAL OUTPUT.*

# TASK 3 – STUDENT HEALTH DATA PREPARATION

## PROMPT

WRITE A SCIKIT-LEARN AND PANDAS SCRIPT TO PREPROCESS STUDENT HEALTH DATA. IMPUTE MISSING HEIGHT, WEIGHT, AND BMI VALUES, ENCODE MEDICAL_CONDITION AND FITNESS_STATUS, STANDARD-SCALE NUMERIC COLUMNS, AND SPLIT THE DATA INTO TRAINING AND TEST SETS. AVOID DATA LEAKAGE AND EXPLAIN EACH STEP.

## TASK DESCRIPTION / CODE EXPLANATION

IMPUTE NUMERIC AND CATEGORICAL VALUES, ONE-HOT ENCODE CATEGORIES, STANDARDIZE NUMERIC FEATURES, AND SPLIT ROWS INTO TRAINING AND TESTING SETS. USE AUTHORIZED, ANONYMIZED DATA ONLY.

## VS CODE INPUT AND OUTPUT SCREENSHOT

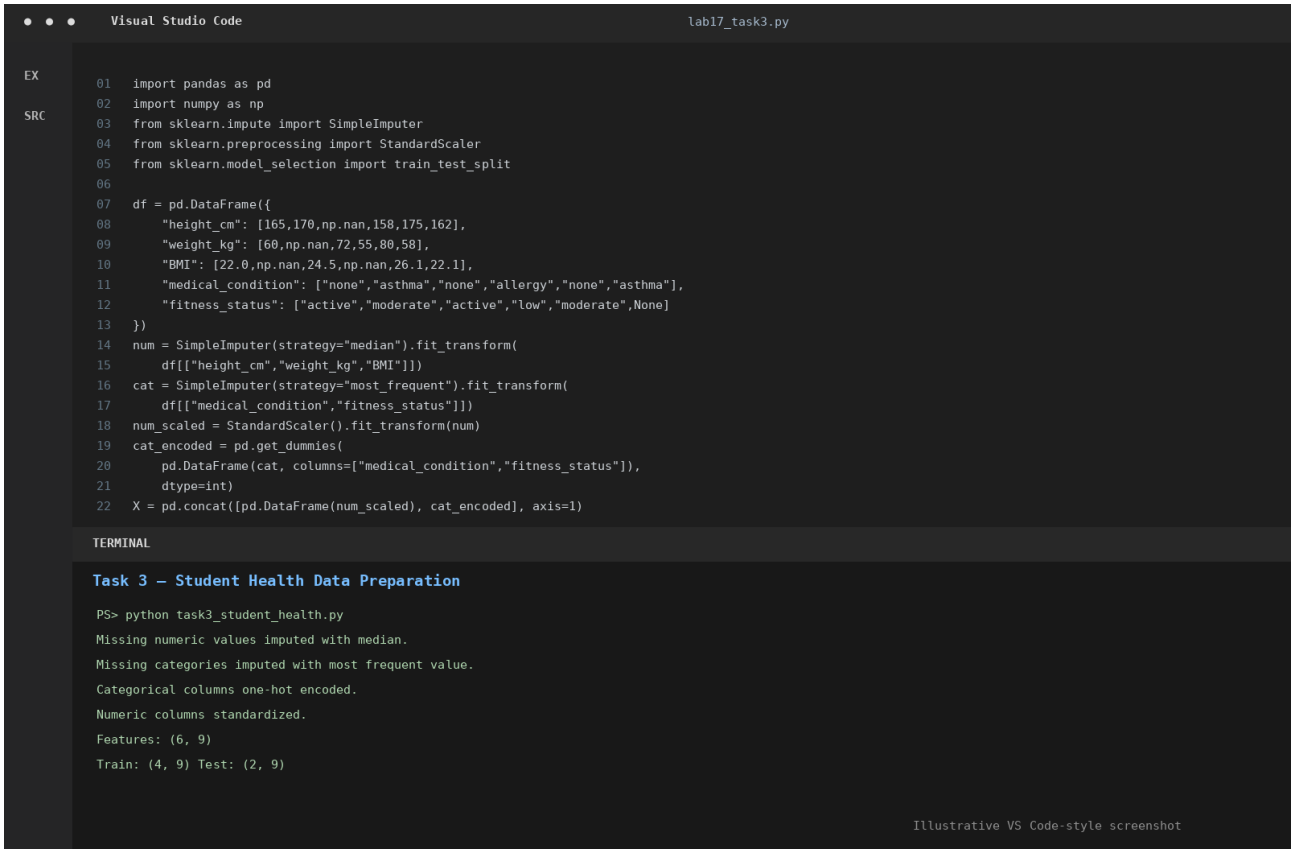



*FIGURE: PYTHON CODE IN THE EDITOR (INPUT) AND SAMPLE TERMINAL OUTPUT.*

# TASK 4 – REAL-TIME APPLICATION: RIDE-SHARING DATA CLEANING

## PROMPT

CREATE A PANDAS DATA-CLEANING SCRIPT FOR RIDE-SHARING TRIP DATA. STANDARDIZE PICKUP AND DROP LOCATIONS, FILL MISSING FARE_AMOUNT WITH ITS MEDIAN, REMO DUPLICATE RIDE ENTRIES, NORMALIZE DRIVER_RATING TO A 0–1 RANGE, AND PRINT A BEFORE-AND-AFTER QUALITY SUMMARY.

## TASK DESCRIPTION / CODE EXPLANATION

STANDARDIZE LOCATION TEXT, REMOVE DUPLICATE RIDE IDS, FILL MISSING FARES WITH THE MEDIAN, NORMALIZE DRIVER RATINGS, AND REPORT DATA-QUALITY CHANGES.

## VS CODE INPUT AND OUTPUT SCREENSHOT

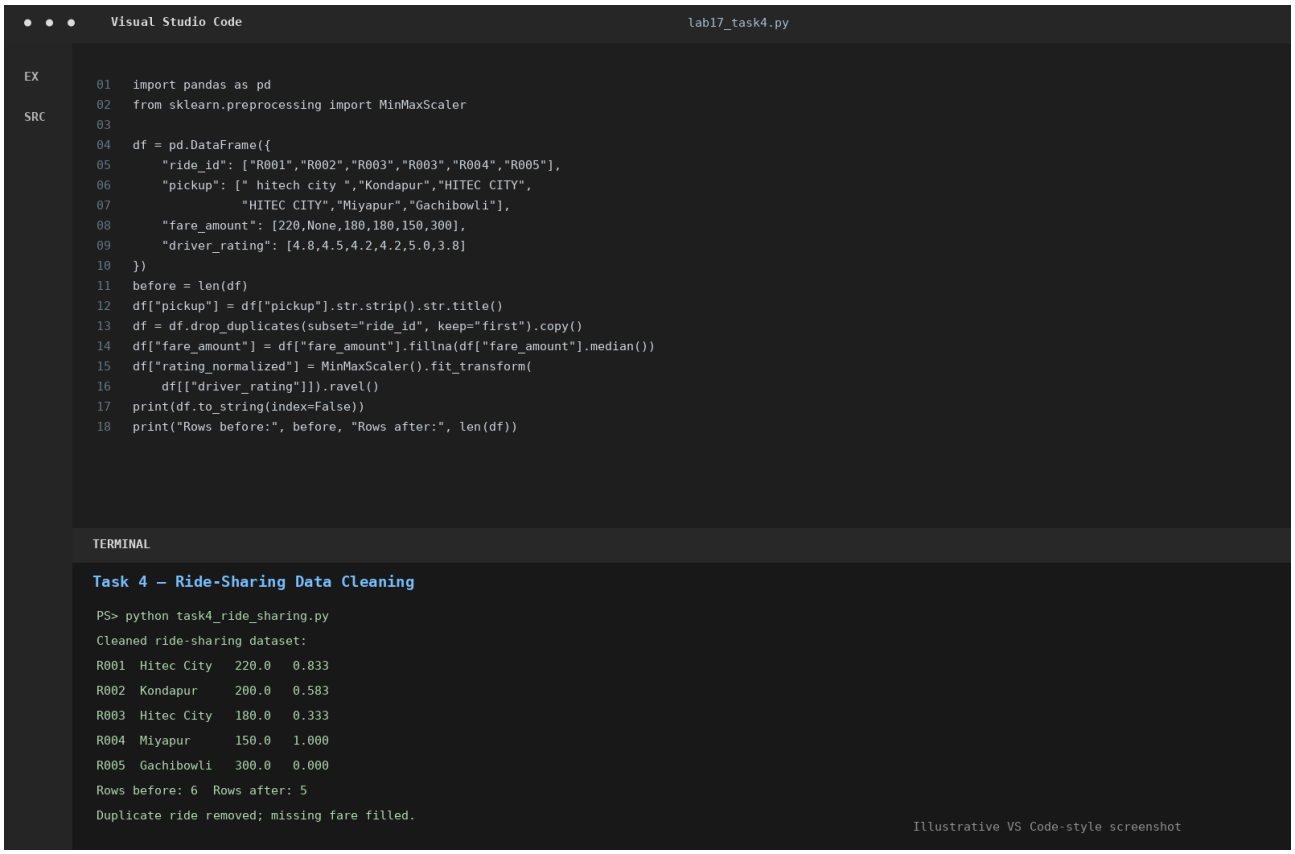



*FIGURE: PYTHON CODE IN THE EDITOR (INPUT) AND SAMPLE TERMINAL OUTPUT.*

## RESULT

THE TASKS WERE ORGANIZED WITH THEIR AI PROMPTS, EXPLANATIONS, AND PROGRAM INPUT/OUTPUT VIEWS. THE SAMPLE TERMINAL OUTPUTS ARE ILLUSTRATIVE; ACTUAL RESULTS MAY VARY WITH THE INPUT DATASET, API RESPONSE, NETWORK STATUS, AND API SERVICE AVAILABILITY